# Microsoft plans to ship a full Linux kernel in Windows 10!

Microsoft has yet another surprise for the Linux developer community. It is planning to ship a full Linux kernel directly in Windows 10. This is the first time that the Linux kernel will be included as part of Windows. The first release will be based on the latest stable Linux release – version 4.19.


"Beginning with Windows Insiders builds this summer, we will include an in-house custom-built Linux kernel to underpin the newest version of the Windows Subsystem for Linux (WSL). This marks the first time that the Linux kernel will be included as a component in Windows," Jack Hammons, program manager, Linux Systems Group, Microsoft, wrote in a blog post.

Hammons explained that like WSL1, WSL2 will not provide any user space binaries. Instead, the Microsoft kernel will interface with a user space installed via the Windows Store. But it can also be 'sideloaded' through the creation of a custom distribution package.

"The kernel itself will initially be based on version 4.19, the latest long-term stable release of Linux. It will be rebased at the designation of new long-term stable releases to ensure that the WSL kernel always has the latest Linux goodness," Hammons said.

The WSL kernel will be built using Microsoft's world-class CI/CD systems and serviced through Windows Update. The company promises to update the kernel with the newest features and fixes.

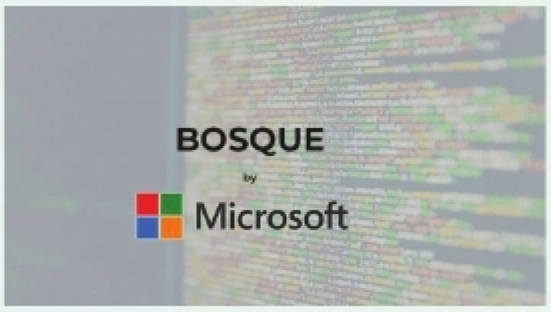
Hammons added, "One of the great things about Linux is its stable and backwards-compatible system call interface. This will enable us to ship the latest stable branch of the Linux kernel to all versions of WSL2. We will rebase the kernel when a new LTS is established and when we have sufficiently validated it."

The kernel provided for WSL2 will be fully open source, which means developers can create their own WSL kernel and contribute changes. When WSL2 is released in Windows Insider builds, instructions for creating your own WSL kernel will be made available on GitHub, say sources at Microsoft.

## Microsoft launches an open source programming language and calls it Bosque

Microsoft has introduced a new open source programming language called Bosque. It's inspired by the syntax of TypeScript, the semantics of machine learning and by Node.js. The main purpose behind Bosque is to build a functional programming language that will help the programmers to move beyond the paradigm of structured programming.

"The Bosque programming language is designed for writing code that is simple, obvious, and easy to reason about, for both humans and machines," said computer scientist Mark Marron, the developer of Bosque. "The key design features of the language provide ways to avoid accidental complexity in the development and coding process. The goal is improved developer productivity, increased software quality, and enabling a range of new compilers and developer tooling experiences," Marron added.




In Bosque, functional programming is combined with block scopes and braces by allowing multiple assignments to updatable variables. Algebraic operations are available for data types, tuples, records, and nominal types, and also for operations that include projection, multi-update and merge. Bulk algebraic data operations begin with bulk reads and data values updates in Bosque.

Bosque comes with first-class support for expressing a range of invariants, diagnostic assertions, and sanity checks. It is in the development stage and experts don't recommend using it for any kind of production work. However, it is open to experiment with.